

Python Full Stack Development Bootcamp - 45 Day Plan

Days 1–35: Teaching Phase

Days 1–14: Python Programming

Day 1: Python Introduction, Installation, Variables, Data Types
Day 2: Operators, Input/Output, Type Casting
Day 3: Conditional Statements
Day 4: Loops
Day 5: Functions
Day 6: Strings
Day 7: Lists
Day 8: Tuples, Sets, Dictionaries
Day 9: OOP Introduction
Day 10: Inheritance
Day 11: Polymorphism
Day 12: Encapsulation, Abstraction
Day 13: Exception Handling, File Handling
Day 14: Modules, Packages, Decorators, Generators
Assignment: Student Management System

Days 15–20: Database (PostgreSQL)

Day 15: DBMS Concepts, PostgreSQL Setup
Day 16: CRUD Operations
Day 17: Constraints
Day 18: Joins
Day 19: Subqueries, Views
Day 20: Indexes, Transactions

Days 21–27: Django

Day 21: Django Architecture, Project Setup
Day 22: Apps, URL Mapping
Day 23: Views
Day 24: Models
Day 25: ORM
Day 26: Forms, Templates
Day 27: Django Admin

Days 28–31: Django REST Framework

Day 28: REST APIs, HTTP Methods
Day 29: Serializers
Day 30: CRUD APIs
Day 31: Authentication, JWT

Days 32–35: Frontend + React

Day 32: HTML, CSS
Day 33: JavaScript
Day 34: React Basics
Day 35: Hooks, Routing, API Integration

Days 36–45: Capstone Project - Hospital Management System

Day 36: Requirement Gathering, Architecture, ER Diagram
Day 37: Database Design (Patients, Doctors, Appointments, Prescriptions, Billing)

Day 38: Django Models & Relationships
Day 39: Authentication & Role-Based Access
Day 40: Patient Management APIs
Day 41: Appointment & Prescription APIs
Day 42: React Setup & Routing
Day 43: Patient, Doctor, Appointment Screens
Day 44: API Integration, Testing, Bug Fixes
Day 45: Deployment (GitHub, Docker, AWS), Resume & Interview Guidance

Hospital Management System Features

- Patient Registration
- Doctor Management
- Appointment Scheduling
- Prescription Management
- Billing Module
- Admin Dashboard
- JWT Authentication
- React Frontend
- Django REST APIs
- PostgreSQL Database

Final Deliverables

Python, PostgreSQL, Django, DRF, React, JWT Authentication, GitHub, Docker, AWS Deployment, and a Production-Ready Hospital Management System.